CO4- AT2 - LAB PRACTICE

Java Lab Practice – Applets, AWT Components and Layout Managers

Question 1-Applet Life Cycle

JAVA CODE:

```
import java.applet.Applet;
import java.awt.Graphics;

public class LifeCycle extends Applet {

    public void init() {
        System.out.println("init() called");
    }

    public void start() {
        System.out.println("start() called");
    }

    public void paint(Graphics g) {
        g.drawString("Applet Life Cycle", 50, 50);
    }

    public void stop() {
        System.out.println("stop() called");
    }

    public void destroy() {
        System.out.println("destroy() called");
    }
}
```

HTML CODE:

```
<html>
<body>

<applet code="LifeCycle.class" width="400" height="200">
</applet>

</body>
</html>
```

OUTPUT:

```
D:\>appletviewer lifecycle.html
init() called
start() called
stop() called
destroy() called

D:\>
```

Question 2-Applet Graphics

JAVA CODE:

```
import java.applet.Applet;
import java.awt.Color;
import java.awt.Graphics;

public class SceneApplet extends Applet {
    public void paint(Graphics g) {

        g.setColor(Color.CYAN);
        g.fillRect(0, 0, 600, 400);

        g.setColor(Color.YELLOW);
        g.fillOval(450, 30, 80, 80);

        g.setColor(Color.WHITE);
        g.fillOval(80, 50, 80, 40);
        g.fillOval(120, 30, 80, 60);
        g.fillOval(160, 50, 80, 40);

        g.setColor(Color.GRAY);
        g.fillRect(0, 300, 600, 100);

        g.setColor(Color.WHITE);
        g.fillRect(50, 345, 80, 10);

        g.fillRect(200, 345, 80, 10);
        g.fillRect(350, 345, 80, 10);
        g.fillRect(500, 345, 80, 10);

        g.setColor(Color.ORANGE);
        g.fillRect(200, 170, 200, 130);
        g.setColor(Color.BLACK);
        g.drawRect(200, 170, 200, 130);
```

```
        g.fillRect(200, 345, 80, 10);
        g.fillRect(350, 345, 80, 10);
        g.fillRect(500, 345, 80, 10);

        g.setColor(Color.ORANGE);
        g.fillRect(200, 170, 200, 130);
        g.setColor(Color.BLACK);
        g.drawRect(200, 170, 200, 130);

        g.drawLine(180, 170, 300, 80);
        g.drawLine(300, 80, 420, 170);
        g.drawLine(180, 170, 420, 170);

        g.setColor(Color.DARK_GRAY);
        g.fillRect(275, 230, 50, 70);

        g.setColor(Color.BLUE);
        g.fillRect(220, 200, 40, 40);
        g.fillRect(340, 200, 40, 40);

        g.setColor(Color.DARK_GRAY);
        g.fillRect(70, 210, 30, 90);

        g.setColor(Color.GREEN);
        g.fillOval(35, 150, 90, 90);
        g.fillOval(60, 130, 90, 90);

        g.setColor(Color.DARK_GRAY);
        g.fillRect(480, 210, 30, 90);

        g.setColor(Color.GREEN);
        g.fillOval(445, 150, 90, 90);
        g.fillOval(470, 130, 90, 90);
    }
}
```

HTML CODE:

```
<html>
<body>

<applet code="SceneApplet.class" width="600" height="400">
</applet>

</body>
</html>
```

OUTPUT:

Question 3-Applet Calculator

JAVA CODE:

```java
import java.applet.Applet;
import java.awt.Graphics;

public class CalculatorApplet extends Applet {

    public void paint(Graphics g) {

        double num1 = Double.parseDouble(getParameter("num1"));
        double num2 = Double.parseDouble(getParameter("num2"));
        String operator = getParameter("operator");

        double result = 0;

        if (operator.equals("+"))
            result = num1 + num2;
        else if (operator.equals("-"))
            result = num1 - num2;
        else if (operator.equals("*"))
            result = num1 * num2;
        else if (operator.equals("/"))
            result = num1 / num2;

        g.drawString("Number 1 = " + num1, 50, 50);
        g.drawString("Number 2 = " + num2, 50, 80);
        g.drawString("Operator = " + operator, 50, 110);
        g.drawString("Result = " + result, 50, 140);
    }
}
```

HTML CODE:

```html
<html>
<body>

<applet code="CalculatorApplet.class" width="400" height="200">
    <param name="num1" value="20">
    <param name="num2" value="10">
    <param name="operator" value="+">
</applet>

</body>
</html>
```

OUTPUT:

Number 1 = 20.0

Number 2 = 10.0

Operator = +

Result = 30.0

Question 4 -Student Profile Using Awt Components

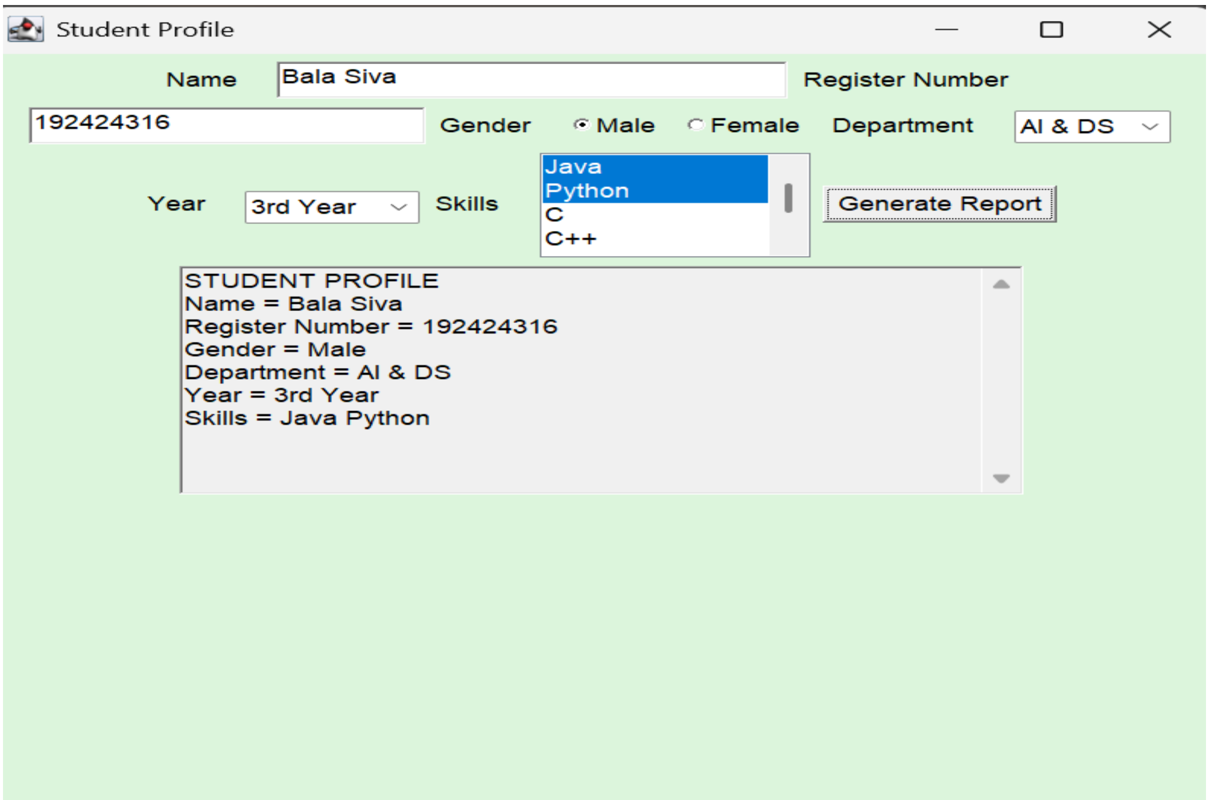
CODE:

```
import java.awt.*;
import java.awt.event.*;

public class StudentProfile extends Frame implements ActionListener
{
    Label L1,L2,L3,L4,L5,L6;
    TextField T1,T2;
    TextArea TA2;
    Choice C1,C2;
    Checkbox male,female;
    CheckboxGroup CG;
    List skills;
    Button B1;

    StudentProfile()
    {
        Frame F1 = new Frame("Student Profile");

        F1.setLayout(new FlowLayout());
        F1.setBackground(new Color(220,245,220));

        L1 = new Label("Name");
        T1 = new TextField(20);

        L2 = new Label("Register Number");
        T2 = new TextField(15);

        L3 = new Label("Gender");
        CG = new CheckboxGroup();

        male = new Checkbox("Male",CG,true);
        female = new Checkbox("Female",CG,false);

        L4 = new Label("Department");
        C1 = new Choice();
        C1.add("AI & DS");
```

```
        C1.add("CSE");
        C1.add("ECE");
        C1.add("EEE");
        C1.add("IT");

        L5 = new Label("Year");
        C2 = new Choice();
        C2.add("1st Year");
        C2.add("2nd Year");
        C2.add("3rd Year");
        C2.add("4th Year");

        L6 = new Label("Skills");

        skills = new List(4,true);
        skills.add("Java");
        skills.add("Python");
        skills.add("C");
        skills.add("C++");
        skills.add("HTML");
        skills.add("SQL");

        B1 = new Button("Generate Report");

        TA2 = new TextArea(8,35);
        TA2.setEditable(false);

        F1.add(L1);
        F1.add(T1);

        F1.add(L2);
        F1.add(T2);

        F1.add(L3);
        F1.add(male);
        F1.add(female);

        F1.add(L4);
        F1.add(C1);

        F1.add(L5);
        F1.add(C2);

        F1.add(L6);
        F1.add(skills);

        F1.add(B1);
        F1.add(TA2);

        B1.addActionListener(this);

        F1.addWindowListener(new WindowAdapter()
        {
            public void windowClosing(WindowEvent e)
            {
                System.exit(0);
            }
        });

        F1.setSize(550,500);
        F1.setVisible(true);
```

```
    }

    public void actionPerformed(ActionEvent ae)
    {
        String skill = "";

        for(String s : skills.getSelectedItems())
        {
            skill = skill + s + " ";
        }

        TA2.setText("");

        TA2.append("STUDENT PROFILE\n");
        TA2.append("Name = " + T1.getText() + "\n");
        TA2.append("Register Number = " + T2.getText() + "\n");
        TA2.append("Gender = " +
                CG.getSelectedCheckbox().getLabel() + "\n");
        TA2.append("Department = " +
                C1.getSelectedItem() + "\n");
        TA2.append("Year = " +
                C2.getSelectedItem() + "\n");
        TA2.append("Skills = " + skill);
    }

    public static void main(String args[])
    {
        new StudentProfile();
    }
}
```

OUTPUT:



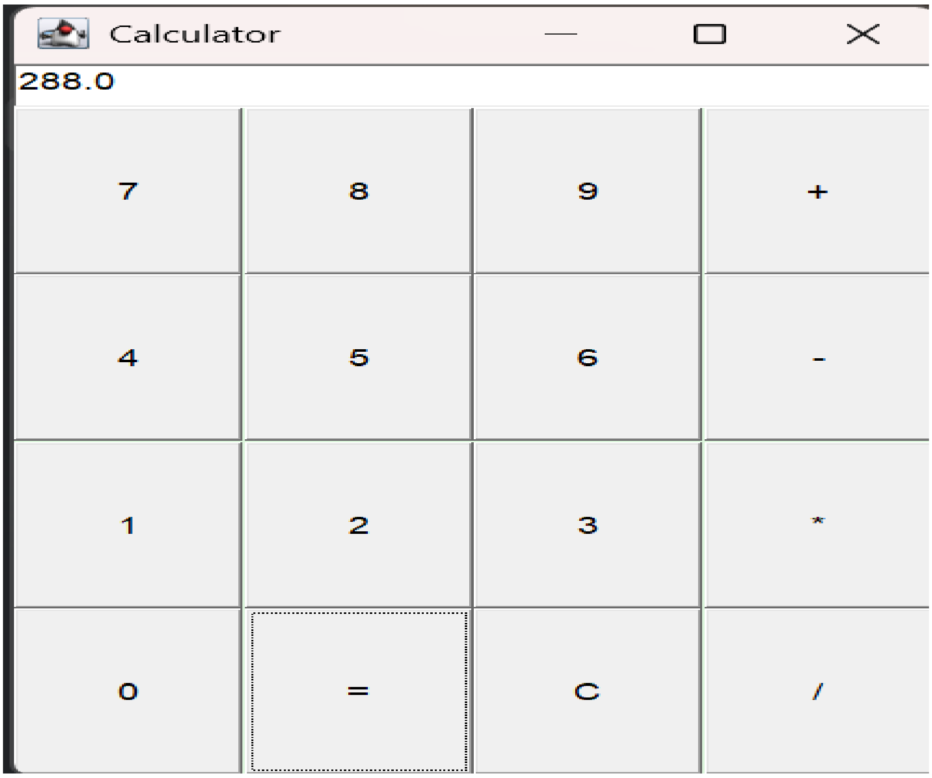
Question 5 -GridLayout with Event Handling

CODE:

```
import java.awt.*;
import java.awt.event.*;

public class GridCalculator extends Frame implements ActionListener
{
    TextField T;
    double n1;
    char op;

    GridCalculator()
    {
        Frame F = new Frame("Calculator");
        F.setLayout(new BorderLayout());
        F.setBackground(new Color(220,245,220));

        T = new TextField();
        F.add(T,BorderLayout.NORTH);

        Panel P = new Panel(new GridLayout(4,4));

        String[] b = {"7","8","9","+","4","5","6","-",
                "1","2","3","*","0","=","C","/"};

        for(String s : b)
        {
            Button B = new Button(s);
            B.addActionListener(this);
            P.add(B);
        }

        F.add(P,BorderLayout.CENTER);

        F.addWindowListener(new WindowAdapter()
        {
            public void windowClosing(WindowEvent e)
            {
                System.exit(0);
            }
        });

        F.setSize(300,400);
        F.setVisible(true);
    }

    public void actionPerformed(ActionEvent e)
    {
        String s = e.getActionCommand();

        if(s.equals("C"))
            T.setText("");

        else if(s.equals("+") || s.equals("-") ||
                s.equals("*") || s.equals("/"))
        {
            n1 = Double.parseDouble(T.getText());
```

```
            op = s.charAt(0);
            T.setText("");
        }

        else if(s.equals("="))
        {
            double n2 = Double.parseDouble(T.getText());

            if(op == '+') n1 += n2;
            if(op == '-') n1 -= n2;
            if(op == '*') n1 *= n2;
            if(op == '/') n1 /= n2;

            T.setText("" + n1);
        }

        else
            T.setText(T.getText() + s);
    }

    public static void main(String[] args)
    {
        new GridCalculator();
    }
}
```

OUTPUT:



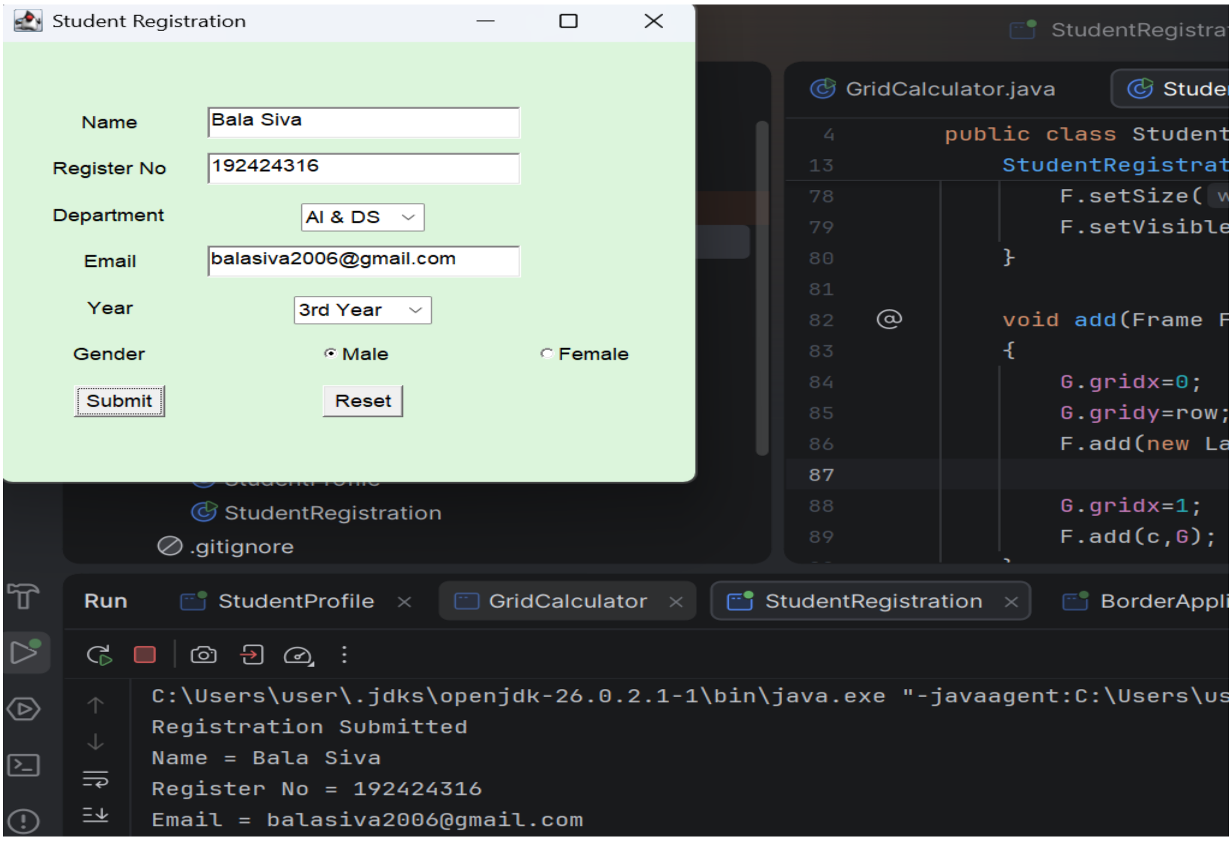
Question 6 – GridBagLayout with Event Handling

CODE:

```
import java.awt.*;
import java.awt.event.*;

public class StudentRegistration extends Frame implements ActionListener
{
    TextField name,reg,email;
    Choice dept,year;
    Checkbox male,female;
    CheckboxGroup gender;
    Button submit,reset;
    GridBagConstraints G;

    StudentRegistration()
    {
        Frame F = new Frame("Student Registration");
        F.setLayout(new GridBagLayout());
        F.setBackground(new Color(220,245,220));

        G = new GridBagConstraints();
        G.insets = new Insets(5,5,5,5);

        name = new TextField(15);
        reg = new TextField(15);
        email = new TextField(15);

        dept = new Choice();
        dept.add("AI & DS");
        dept.add("CSE");
        dept.add("ECE");

        year = new Choice();
        year.add("1st Year");
        year.add("2nd Year");
        year.add("3rd Year");
        year.add("4th Year");

        gender = new CheckboxGroup();
        male = new Checkbox("Male",gender,false);
        female = new Checkbox("Female",gender,false);

        submit = new Button("Submit");
        reset = new Button("Reset");

        add(F,"Name",name,0);
        add(F,"Register No",reg,1);
        add(F,"Department",dept,2);
        add(F,"Email",email,3);
        add(F,"Year",year,4);

        G.gridx=0;
        G.gridy=5;
        F.add(new Label("Gender"),G);

        G.gridx=1;
        F.add(male,G);
```

```java
        G.gridx=2;
        F.add(female,G);

        G.gridx=0;
        G.gridy=6;
        F.add(submit,G);

        G.gridx=1;
        F.add(reset,G);

        submit.addActionListener(this);
        reset.addActionListener(this);

        F.addWindowListener(new WindowAdapter()
        {
            public void windowClosing(WindowEvent e)
            {
                System.exit(0);
            }
        });

        F.setSize(400,350);
        F.setVisible(true);
    }

    void add(Frame F,String s,Component c,int row)
    {
        G.gridx=0;
        G.gridy=row;
        F.add(new Label(s),G);

        G.gridx=1;
        F.add(c,G);
    }

    public void actionPerformed(ActionEvent e)
    {
        if(e.getSource() == submit)
        {
            System.out.println("Registration Submitted");
            System.out.println("Name = " + name.getText());
            System.out.println("Register No = " + reg.getText());
            System.out.println("Email = " + email.getText());
        }
        else
        {
            name.setText("");
            reg.setText("");
            email.setText("");
        }
    }

    public static void main(String[] args)
    {
        new StudentRegistration();
    }
}
```

OUTPUT:



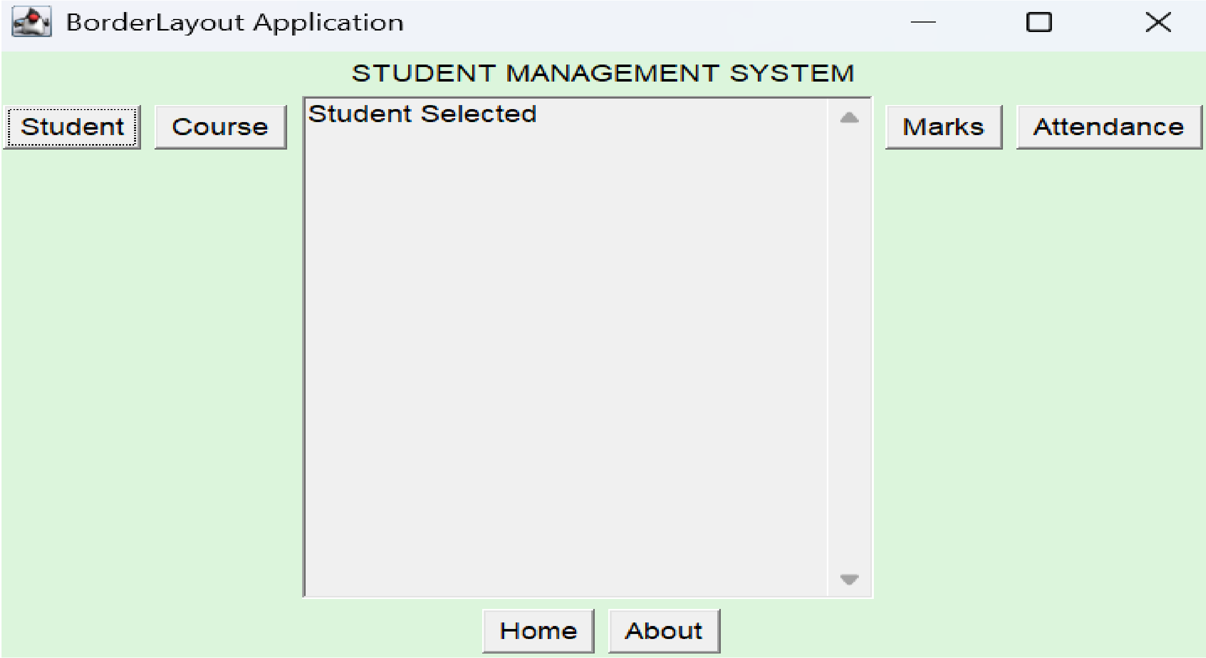
Question 7 -BorderLayout with Event Handling

CODE:

```
import java.awt.*;
import java.awt.event.*;

public class BorderApplication extends Frame implements ActionListener
{
    TextArea T;

    BorderApplication()
    {
        Frame F = new Frame("BorderLayout Application");
        F.setLayout(new BorderLayout());
        F.setBackground(new Color(220,245,220));

        F.add(new Label("STUDENT MANAGEMENT SYSTEM",
                Label.CENTER),BorderLayout.NORTH);

        Panel south = new Panel();

        Button home = new Button("Home");
        Button about = new Button("About");

        south.add(home);
        south.add(about);
```

```java
        Panel east = new Panel();

        Button marks = new Button("Marks");
        Button attendance = new Button("Attendance");

        east.add(marks);
        east.add(attendance);

        Panel west = new Panel();

        Button student = new Button("Student");
        Button course = new Button("Course");

        west.add(student);
        west.add(course);

        T = new TextArea();
        T.setEditable(false);

        F.add(south,BorderLayout.SOUTH);
        F.add(east,BorderLayout.EAST);
        F.add(west,BorderLayout.WEST);
        F.add(T,BorderLayout.CENTER);

        home.addActionListener(this);
        about.addActionListener(this);
        marks.addActionListener(this);
        attendance.addActionListener(this);
        student.addActionListener(this);
        course.addActionListener(this);

        F.addWindowListener(new WindowAdapter()
        {
            public void windowClosing(WindowEvent e)
            {
                System.exit(0);
            }
        });

        F.setSize(500,350);
        F.setVisible(true);
    }

    public void actionPerformed(ActionEvent e)
    {
        T.setText(e.getActionCommand() + " Selected");
    }

    public static void main(String[] args)
    {
        new BorderApplication();
    }
}
```

OUTPUT:



Question 8 – CardLayout with Event Handling

CODE:

```
import java.awt.*;
import java.awt.event.*;

public class CardApplication extends Frame implements ActionListener
{
    CardLayout card;
    Panel P;
    TextField name,reg,dept,year;
    TextArea report;
    Button next,previous,submit;

    CardApplication()
    {
        Frame F = new Frame("Student Information");

        card = new CardLayout();
        P = new Panel(card);

        Panel C1 = new Panel();

        C1.add(new Label("Student Information"));
        C1.add(new Label("Name"));
        name = new TextField(15);
        C1.add(name);
        C1.add(new Label("Register Number"));
        reg = new TextField(15);
        C1.add(reg);

        Panel C2 = new Panel();
```

```
        C2.add(new Label("Academic Information"));
        C2.add(new Label("Department"));
        dept = new TextField(15);
        C2.add(dept);
        C2.add(new Label("Year"));
        year = new TextField(15);
        C2.add(year);

        Panel C3 = new Panel();

        C3.add(new Label("Confirmation / Report"));
        report = new TextArea(8,30);
        report.setEditable(false);
        C3.add(report);

        P.add(C1,"Card1");
        P.add(C2,"Card2");
        P.add(C3,"Card3");

        Panel B = new Panel();

        previous = new Button("Previous");
        next = new Button("Next");
        submit = new Button("Submit");

        B.add(previous);
        B.add(next);
        B.add(submit);

        previous.addActionListener(this);
        next.addActionListener(this);
        submit.addActionListener(this);

        F.add(P,BorderLayout.CENTER);
        F.add(B,BorderLayout.SOUTH);

        F.addWindowListener(new WindowAdapter()
        {
            public void windowClosing(WindowEvent e)
            {
                System.exit(0);
            }
        });

        F.setSize(500,350);
        F.setVisible(true);
    }

    public void actionPerformed(ActionEvent e)
    {
        if(e.getSource() == next)
            card.next(P);

        else if(e.getSource() == previous)
            card.previous(P);

        else if(e.getSource() == submit)
        {
            report.setText(
                    "STUDENT REPORT\n\n" +
```

```
                                "Name = " + name.getText() + "\n" +
                                "Register Number = " + reg.getText() + "\n" +
                                "Department = " + dept.getText() + "\n" +
                                "Year = " + year.getText()
                );

                card.show(P,"Card3");
            }
        }

    public static void main(String[] args)
    {
        new CardApplication();
    }
}
```

OUTPUT: